8. (a) The photographs show three different methods of fighting fires.

Each method removes a different part of the fire triangle.

Method A

Method B

Method C

Examiner only

Use the fire triangle to explain how the different methods shown in the photographs help to extinguish fires or prevent them from spreading. [6 QER]

(b) **Fighting Fires**

There are five main types of fire extinguisher – water, foam, dry powder, carbon dioxide and wet chemical. Modern fire extinguishers have a red body with a coloured band at the top. Each colour represents a different type of extinguisher, used on different types or 'classes' of fire.

There is no one extinguisher type that works on all classes of fire.

The chart below shows which type of extinguisher should be used on each class of fire.

TYPE ▶ FIRE ▼ EXTINGUISHER	CLASS A Flammable solids, for example paper	CLASS B Flammable liquids, for example ethanol	CLASS C Flammable gases, for example methane	CLASS D Flammable metals, for example magnesium	CLASS E Electrical equipment, for example computer	CLASS F Cooking oil, for example deep fat fryer
WATER	✓	✗	✗	✗	✗	✗
FOAM	✓	✓	✗	✗	✗	✗
DRY POWDER	✓	✓	✓	✓	✓	✗
CO_2	✗	✓	✗	✗	✓	✗
WET CHEMICAL	✓	✗	✗	✗	✗	✓

Examiner only

Use only the information in the chart to answer parts (i)–(iii).

(i) Underline the type of extinguisher used to put out a fire involving burning cooking oil. [1]

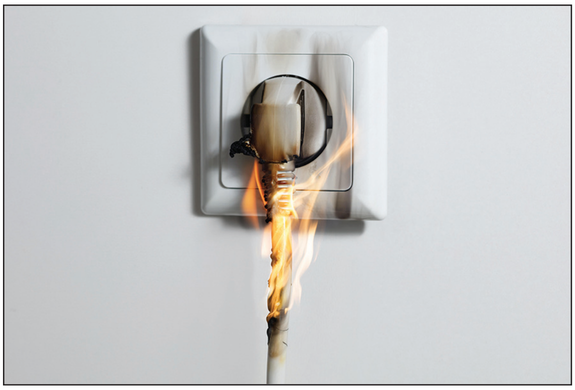
water **foam** **dry powder** **carbon dioxide** **wet chemical**

(ii) Underline the type of extinguisher that would be the most useful in a school chemistry laboratory. [1]

water **foam** **dry powder** **carbon dioxide** **wet chemical**

(iii) Tick (✓) the box next to the fire that should be put out using a water extinguisher. [1]

chip pan fire ☐



burning plug and socket ☐

burning butane cylinder ☐

burning waste cardboard ☐

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